Early Detection of Forest Disturbances in Homogeneous Stands – Deep Learning-Based Monitoring of *Picea Abies*

Maximilian Kirsch, Jakob Wernicke, Pawanjeet Singh Datta, and Christine Preisach

***Abstract*—Climate change has increased the vulnerability of forests to insect-related damage, resulting in widespread forest loss in Central Europe and highlighting the need for effective, continuous monitoring systems. Remote sensing-based approaches, often rely on supervised machine learning (ML) algorithms that require labeled training data. Monitoring temporal patterns through time series analysis offers a potential alternative for earlier detection of disturbances but requires substantial storage resources. This study investigates the potential of a Deep Learning algorithm based on a Long Short Term Memory (LSTM) Autoencoder for the detection of anomalies in forest health (e.g., bark beetle outbreaks), utilizing Sentinel-2 time series data. This approach is an alternative to supervised ML methods, avoiding the necessity for labeled training data. Furthermore, it is more memory-efficient than other time series analysis approaches, as a robust model can be created using only a 26-week-long time series as input. In this study, we monitored pure stands of spruce in Thuringia, Germany, over a 7-year period from 2018 to the end of 2024. Our best model achieved a detection accuracy of 87% on test data and was able to detect 65% of all anomalies at a very early stage (more than a month before visible signs of forest disturbance). Compared to the established time series break detection algorithm – breaks for additive season and trend and isolation forest anomaly detection, our approach consistently detected a higher percentage of anomalies at an earlier stage. These findings suggest that LSTM-based Autoencoders could provide a promising, resource-efficient approach to forest health monitoring, enabling more timely responses to emerging threats.**

***Index Terms*—Anomaly detection, autoencoder, bark beetle, forest, long short term memory (LSTM), remote sensing, spruce.**

I. Introduction

FORESTS in Central Europe are increasingly threatened by intensified insect outbreaks, largely driven by climate change. In Thuringia, Germany, for example, approximately 20% of the forest area was affected by a bark beetle infestation between 2018 and 2023, eventually resulting in a timber loss of 28 million m^3 in the region [1]. The primary impact was observed in pure stands of Norway spruce (*Picea abies*), which were predominantly affected by the two most common bark beetle species in Germany: *Ips typographus* and *Pityogenes chalcographus*. This level of forest disturbance is unprecedented not only in Central Germany, but also across many spruce-dominated forest regions throughout Central Europe. Climate change induces stressors that negatively impact the overall health and growth of forests [2], [3]. Particularly, hot summers and associated droughts impair the efficacy of defense mechanisms of spruce trees [4]. Secondary pests like thermophilic bark beetle populations thrive under such circumstances, as their reproduction depends on predisposed and already weakened trees [5], [6].

Predicting disturbances is challenging due to the unpredictable nature of insect-induced calamities [7]. An early and spatially precise detection of potential bark beetle infestation centers are needed to mitigate an epidemic bark beetle reproduction [8], [9]. Accordingly, several studies stressed the hypothesis of green attack detection based on remote sensing data [10], [11], [12]. However, there is wide consensus that green attack detection is either unreliable or infeasible on large spatial scales [9], [13], [14]. Instead, our focus is on identifying anomalous patterns in forest health that may serve as early warning indicators of potential disturbances. A limitation of early disturbance detection is often the lack of labeled training data. Satellite image time series (SITS) based anomaly detection can operate unsupervised and does not rely on labeled training samples to capture seasonal variations [15], [16]. Often SITS approaches, such as the Breaks for Additive Season and Trend (BFAST) algorithm, learn temporal patterns, and detect deviations from the normal state [17]. Other approaches utilize deep learning such as LSTM neural networks that remember temporal patterns in time series [18]. Most approaches depend on extensive historical information for every observed time series and often monitor univariate time series, creating hyperspectral relationships only by calculating vegetation indices (VIs) [15], [16], [18].

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We investigated the potential of a deep learning, reconstruction-based time series anomaly detection approach to create an early warning system based on Sentinel-2 satellite images. We built a LSTM-AE model for real-time anomaly detection with a 10-m spatial resolution. We applied our approach to data collected from severely threatened spruce stands in Thuringia, Germany, between 2018 and the end of 2024. Through observation and measurement of the health dynamics of the stands, we aim to flag anomalies in early disturbance stages wall-to-wall. Our algorithm employs unsupervised deep learning and is independent from labeled training data. This makes our method accessible for a wide range of users, especially paired with freely available Sentinel-2 data. In addition, unlike many SITS approaches, this technique minimizes the need for extensive historical data, which is particularly important for single cell-based analyses in order to reduce memory demands. The implementation of this early warning system is suitable for an operational forest disturbance monitoring and thus might support forest health management strategies.

Our main contributions are as follows.

1) We propose a robust unsupervised approach for SITS anomaly detection in forest health dynamics that is highly accurate in detecting anomalies up to several months earlier than static methods based on VI thresholds, by leveraging the temporal relationship between the multispectral channels on a single cell basis.
2) We introduce an approach for online anomaly detection, leveraging freely available Sentinel-2 satellite data. In contrast to conventional SITS methods, that require extended historical time series, our method processes incoming data in real time, mitigating the need for substantial temporal archives while maintaining robust detection performance.
3) We analyzed the importance of temporal information by testing LSTM-AE models that observe diverse window sizes from one month to one year. Thereby balancing model performance and historical data requirements to reduce storage demands.
4) We compared the LSTM models against a SITS method for break detection in time series called BFAST and the anomaly detection algorithm isolation forest (IF). Our model achieved a higher level of accuracy, as well as a significant increase in the number of correct early predictions.
5) Ultimately, we propose a scoring method that allows us to compare diverse models' anomaly detection and early warning potential exceeding the generally used performance metrics, by rewarding earlier detections.

II. Related Work

Early detection of bark beetle outbreaks is a critical research area due to their significant impact on forest ecosystems. Detecting infested trees during the "green attack" stage – when beetles have infested a tree but crown symptoms are not yet visible – is particularly challenging, even in situ. This stage precedes the "red attack," where canopy discoloration becomes detectable in aerial imagery [14].

A. Remote Sensing-Based Bark Beetle Detection

Classifying green attack using single remote sensing images is considered infeasible with medium-resolution satellite data [14]. Some studies have explored machine learning (ML) approaches to distinguish green attack from healthy vegetation. Abdullah et al. [10] analyzed Sentinel-2 spectral bands and VIs, using statistical ML for variable selection. However, the results were deemed unsuitable for forestry applications by the authors [10]. Other studies focused on changes in forest stress levels using random forest classification, to distinguish bark beetle infested trees from healthy vegetation [19]. These supervised ML approaches rely on labeled training data. The difficulty of creating reliable labels from satellite data, makes supervised ML impractical.

B. SITS Analysis

SITS analysis is widely used to detect forest disturbances, including bark beetle outbreaks [11], [20], [21]. A key advantage of SITS is its ability to retrospectively analyze patterns leading to disturbances, such as red attack. SITS has been applied to classify bark beetle infestations using ML techniques. For instance, Barta et al. [11] used a random forest model on SITS to pinpoint early separability between healthy and infested trees across various indices. They identified Sentinel-2 spectral bands and VIs that best differentiate healthy from infested trees. Abdullah et al. [20] explored the relationship between bark beetle and canopy chlorophyll content derived from SPOT-5 and RapidEye SITS data in combination with ground truth data.

Anomaly detection methods are also prevalent in SITS research. An approach, introduced by Dutrieux et al. [22] proposed a model, that detects time series harmonics and forecasts the pattern and if the actual time series deviates strongly from that pattern, an anomaly is flagged. The well known break detection model BFAST, detects changes by decomposing time series into trend, seasonal, and residual components [17]. This approach is commonly utilized to detect structural changes in vegetation time series [16], [23]. Furthermore, the sequential analysis method CUSUM has been employed to identify deviations in attacked trees from standard SITS patterns [21], [24]. Another unsupervised anomaly detection method is the Isolation Forest (IF) algorithm. IF is frequently applied in time series anomaly detection [25] and has also been successfully used in SITS applications [26].

C. Deep Learning Approaches

Deep learning has gained traction in SITS for forest health classification and disturbance detection [18], [27], [28], [29]. Zhou et al. [28] proposed an unsupervised AE model that detects anomalies in univariate Landsat time series by identifying reconstruction errors. Recurrent Neural Networks (RNNs), particularly using LSTM architectures, excel at capturing temporal dynamics in vegetation phenology and outperform traditional methods in land cover classification [30], [31]. LSTM models have been applied to change detection on Sentinel-1 data [32] and burned area mapping using MODIS data [18], [27]. Kong et al. [18] introduced a LSTM-based framework that forecasts

time series and flags disturbances when predictions deviate significantly from observations.

Furthermore, recent advances in disturbance detection from SITS include the work by Schiller et al. [29], who introduced a transformer-based model for classifying time series as disturbed or undisturbed. Their approach demonstrated the ability to detect even subpixel-level anomalies. However, the model required extensive amounts of labeled training data (their model was trained on 1.5 million pixel time series) and operated on long temporal inputs, relying on four-years of time series for each observation. High-quality labeled data are often limited in forest disturbance detection, and the use of long time series considerably increases storage and computational demands.

In parallel, LSTM-AE have been widely used for unsupervised anomaly detection across domains, such as renewable energy monitoring, medical signal processing, cybersecurity, and environmental systems [33], [34], [35], [36]. These models effectively capture complex temporal dependencies, particularly in multivariate settings. However, to the best of our knowledge, LSTM-AE have not been explored extensively in the context of forest monitoring systems.

Our study explores the use of multivariate LSTM-AE models on Sentinel-2 time series for detecting forest health disturbances in spruce stands. Unlike many existing approaches that rely on univariate inputs, long historical records, or extensive labeling, our method leverages multispectral and temporal patterns over shorter input windows (e.g., 26 to 52 weeks) to identify early signs of disturbance. By evaluating different input lengths and using a reconstruction-based unsupervised framework, we aim to minimize false positives while reducing data volume and computational demands. This work addresses a key gap in current SITS-based forest monitoring by enabling early, label-free disturbance detection from shorter multivariate sequences.

III. Study Area and Data

In this chapter, we describe our study areas, the larger study area, the Thuringian Forest and the Südharz Nature Park, which are the selected areas for training and testing, and the Sentinel-2 time series data basis.

A. Study Area

In order to evaluate the potential of LSTM-AE for early disturbance detection, we observed Norway spruce (*Picea abies*) stands in the Thuringian Forest and the southern part of the Harz Mountains (central Germany) retrospectively during the period from January 2018 to December 2024. Thuringia is a suitable study area for the detection of disturbances due to its extensive bark-beetle induced forest degradation since 2018. The predominately managed forests of Thuringia are composed of three main tree species: Norway spruce, European beech (*Fagus sylvatica*), and Scots pine (*Pinus sylvestris*), which are partly intermixed with secondary tree species. Before 2018, the region was covered primarily by single-layered spruce stands of a uniform age, reflecting the typical forest structure of the time. The persistent drought since 2018 initiated the most severe forest degradation in the region. Consequently, approximately 110 000 ha of primarily spruce-dominated stands were clear-cut due to extensive bark beetle outbreaks, following the prolonged drought since 2018. [1].

We used available open forest mapping data provided from Thüringen Forest[1] to select conifer pure stands. Since no detailed tree species data were available we additionally used true-color, orthorectified aerial imagery (orthophotos with a spatial resolution of 0.2 m), provided by the State of Thuringia and recorded on June 4, 2023.[2] These images were used to visually identify and select areas corresponding to pure spruce stands exhibiting varying degrees of forest health.

We selected four training areas, as well as a larger area for testing in the Thuringia Forest region. In addition, eight test areas were chosen from the Südharz Nature Park, located in the northern part of Thuringia (see Fig. 1). The model was trained using time series of healthy spruce stands. For this, we selected visually healthy forest patches with green crowns as training areas, covering a total of 28.85 ha. The health status of these areas throughout the observation period was confirmed by analyzing the time series of VIs (NDVI, DWSI). The selected training areas are shown in Fig. 2.

The areas designated for testing depict varying degrees of disturbance (e.g., healthy forest, crown discolorations and clear-cuts). The polygonal areas varied in size from 25 to 561 pixels totaling 3175 pixels (31 ha). Furthermore, we selected a larger test region in which various stages of disturbance were represented in a contiguous area of 44 519 pixels (445 ha). We used this region to create an anomaly map.

B. Sentinel-2 Time Series

The Sentinel-2 data were selected for their high spatial resolution, frequent revisit time, and free availability. These data capture high-resolution images of the Earth's surface with a combined revisit interval of five days [37].

We selected atmospherically corrected Sentinel-2 Level 2 A (L2A) data, which also include a scene classification (SCL) map [38]. The SCL algorithm characterizes every pixel in a Sentinel-2 image into one of 12 classes [39]. These classes describe cloud types, snow, vegetation, or nonegetated areas, which we utilized to filter instances where an observed pixel was covered by cloud or snow at a given time step.

The Sentinel-2 L2A data is represented by multivariate time series for each pixel, covering 12 spectral bands from January 2018 to December 2023. These were organized into spatially and temporally indexed tables.

IV. Background: LSTM Autoencoder Model

In this section, we provide a comprehensive overview of the model components and the model architecture of the unsupervised Deep Learning algorithm we used for early detection of bark-beetle outbreaks. First, the general functionality of Autoencoder (AE)'s is set out, followed by the operating principles of

[1]Mapping data: https://geomis.geoportal-th.de/geonetwork/srv/ger/catalog.search#/metadata/c70f0858-af9b-4395-9a0d-1b577e6d09e6. Last accessed on 30 August 2024.

[2]Orthophotos: https://geoportal.thueringen.de/gdi-th/download-offene-geodaten/download-luftbilder-und-orthophotos. Last accessed: 30 August 2024

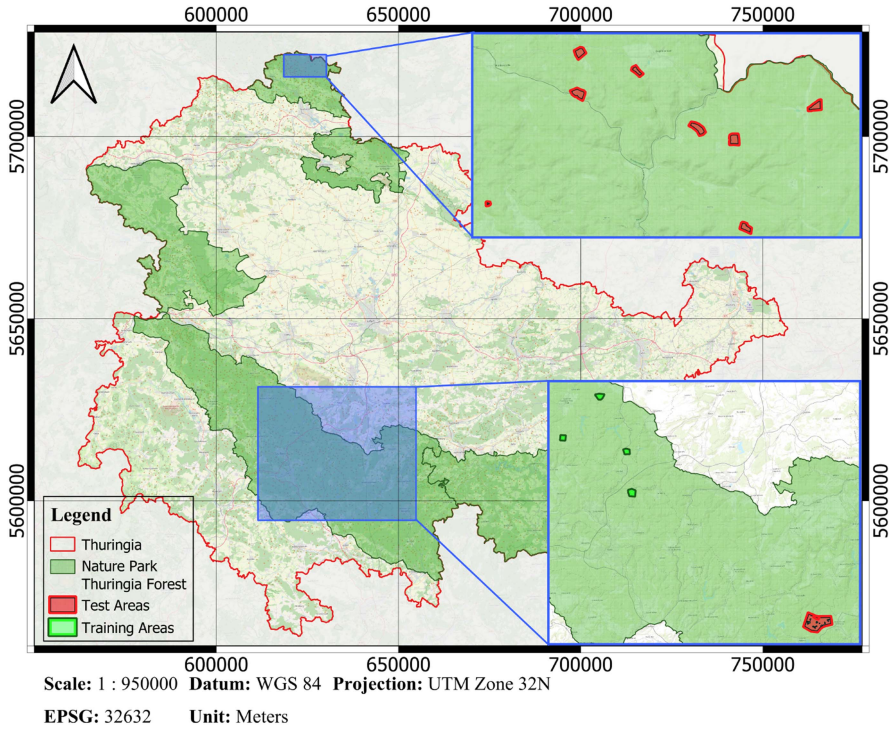


Fig. 1. Locations of the study area in Thuringia, Germany.

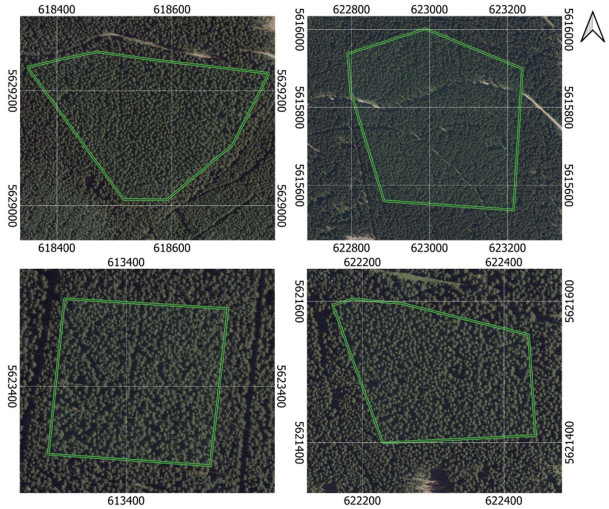


Fig. 2. Areas selected to use as training data. The images show recordings from June 4, 2023.

LSTM. In conclusion, we outline the general functionality and structure of the LSTM AE model.

A. Autoencoder

We built an AE for anomaly detection. An AE is a neural network designed to reconstruct input data using two functions: an encoder and a decoder [35]. The encoder E learns the most important features of the input data x and reduces them into a lower-dimensional representation called the latent space z [40]

$$z = E(x). \tag{1}$$

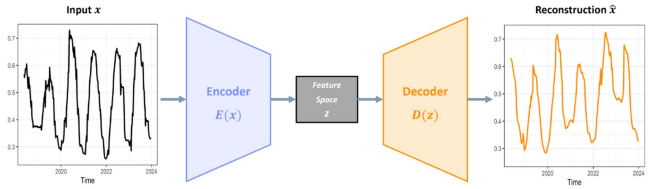


Fig. 3. Autoencoder: Reconstruction of a time series.

The decoder D converts the latent space representation z back to the original shape, producing the reconstructed data $\hat{x}$ [40]

$$\hat{x} = D(z). \tag{2}$$

The structure of an AE and the process of reconstruction is illustrated in Fig. 3.

During training, the AE optimizes parameters, such as weights and biases, to minimize the reconstruction error between the input x and the reconstruction $\hat{x}$ [41]. In anomaly detection, AE's are trained exclusively on "normal" data samples, without anomalies and learn to capture the key features [42], [43]. The difference between input data x and reconstructed data $\hat{x}$ is known as the reconstruction error [41]. Since anomalies exhibit a typical characteristics, they cannot be accurately reconstructed, leading to high reconstruction errors [44]. Anomalies are identified when the reconstruction error exceeds a threshold chosen based on the distribution of reconstruction errors [44].

B. LSTM

We incorporated LSTM into our AE model, making it an effective method for reconstruction-based anomaly detection in

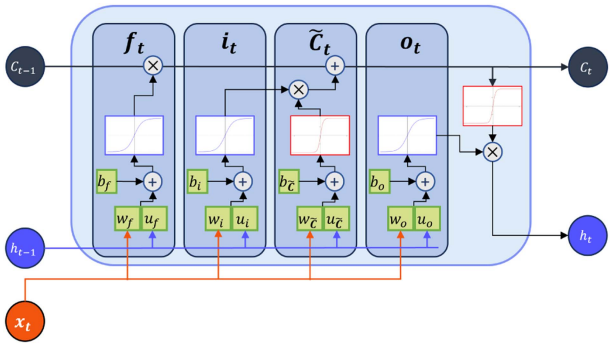


Fig. 4. Components of a LSTM-Unit.

time series [35]. LSTM is a type of RNN, that is designed to efficiently process sequential data e.g., time series. This ensures that temporal relationships are preserved during processing [35]. The LSTM architecture includes a component for maintaining long-term dependencies and another for propagating short-term memories [45]. At each time step, new information is added, updating the memory of the cell [45]. The LSTM unit architecture that we applied is an adaptation of the original LSTM model from [46] and incorporates a mechanism known as the forget gate. This advanced LSTM cell, proposed by [47], adds a weighted function that determines which information from the input should be discarded. The LSTM architecture is illustrated in Fig. 4.

Fig. 4 depicts the components within a basic LSTM cell, as well as the structural flow of the data. The Cell State, denoted C, represents the long-term memory, while the Hidden State, denoted h, stores information about the short-term memory. The input data x_t describes the values for the multivariate time series at the current time step. Furthermore, the LSTM unit contains so called gated units, these are the forget gate f_t, input gate i_t, cell candidate gate $\tilde{C}_t$, and output gate o_t. The weights w, u, and the bias b are learned on the input data and the hidden state, and an activation function σ or $\tanh$ is applied. σ is displayed with a blue frame and $\tanh$ red.

Each LSTM unit has three inputs: 1) The input data of the current time step x_t, 2) the Cell State of the previous time step C_{t-1}, and 3) the Hidden State of the previous time step h_{t-1}. Furthermore, it creates a new Cell State C_t and a new Hidden State h_t as outputs.

The gated units, as well as the information forward pass of an LSTM unit, are defined by the following equations:

$$f_t = \sigma\left(w_f x_t + u_f h_{t-1} + b_f\right) \tag{3}$$

$$i_t = \sigma\left(w_i x_t + u_i h_{t-1} + b_i\right) \tag{4}$$

$$o_t = \sigma\left(w_o x_t + u_o h_{t-1} + b_o\right) \tag{5}$$

$$\tilde{C}_t = \tanh\left(w_{\tilde{C}} x_t + u_{\tilde{C}} h_{t-1} + b_{\tilde{C}}\right) \tag{6}$$

$$C_t = f_t \odot C_{t-1} + i_t \odot \tilde{C}_t \tag{7}$$

$$h_t = o_t \odot \tanh\left(C_t\right) \tag{8}$$

where $\odot$ denotes the point-wise multiplication of two vectors.

The sigmoid activation function is calculated as follows:

$$\sigma(x) = \frac{1}{1 + e^{-x}}. \tag{9}$$

The tanh activation function is calculated as follows:

$$\tanh(x) = \frac{e^{x} - e^{-x}}{e^{x} + e^{-x}}. \tag{10}$$

The operational mechanics of this LSTM unit, as detailed by [48], are as follows:

The equations 3–5 calculate the gated units. Each gated unit vector multiplies the input at the current time step x_t with the respective weights w and multiplies the Hidden State of the previous time step h_{t-1} with the respective recurrent weight u. The vectors are then summed before a bias is added. The results are then passed through a sigmoid activation function, producing a value between 0 and 1. The gated units differ only in terms of their respective weights and biases, allowing the LSTM cell to forget irrelevant information of a sequence [41], [45].

Equation (6), the *cell candidate gate* defines the potential new Cell State $\tilde{C}_t$. Likewise to the equation of the gated units, x_t and h_{t-1} are multiplied with the respective weights w_C and u_C. The products are summed and a bias b_C is then added. The result serves as input for a tanh activation function that regulates the information flow in the network.

To determine the new Cell State C_t, (7) takes the previous Cell State C_{t-1} and multiplies it by the output of the forget gate. Since f_t is a value between 0 and 1, it dictates the proportion of C_{t-1} that is retained. Subsequently, the potential new Cell State $\tilde{C}_t$ is point-wise multiplied by i_t to determine the proportion of $\tilde{C}_t$ that is added.

Equation (8) defines the new Hidden State h_t. It is calculated by using the new Cell State C_t as input for a Tanh Function resulting in a potential new Hidden State. This is then point-wise multiplied by the output gate o_t to determine the proportion of potential Hidden State that is passed on to the next time step.

C. LSTM-AE Architecture

We incorporate stacked LSTM neurons into an AE network (see Fig. 5). LSTM units detect long- and short-term dependencies, as well as relationships between input features, making them highly effective for multivariate time series anomaly detection [35], [40].

We applied a LSTM-AE architecture based on the method proposed by [35]. They built an LSTM AE to detect DDoS attacks[3] in multivariate time series data.

Before applying the model to a time series, the data are divided into sequences of a predefined window size, as explained in Section V-A. Within each sequence, the relationships between features and time steps are computed using hidden LSTM units [35].

Multiple LSTM units are stacked in each layer of the AE. Within each unit, the long-term memory $C_{t,i}$ (where t represents time and i denotes the unit index) and the short-term memory $h_{t,i}$

[3]DDoS stands for Distributed Denial of Service. DDoS attacks are cyberattacks aimed at overwhelming a server by sending enormous amounts of data [35].

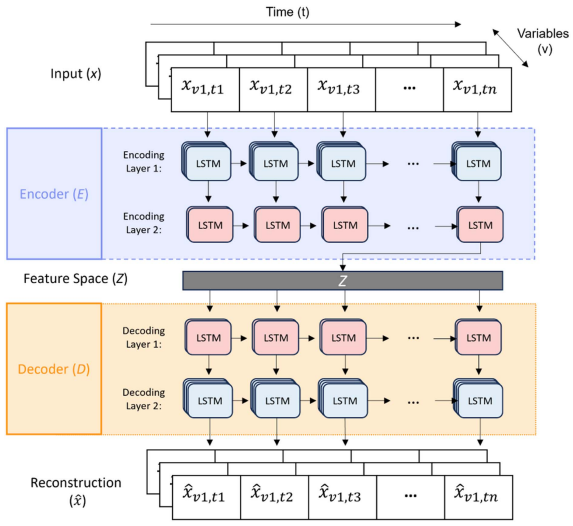


Fig. 5. Conceptual illustration of an LSTM-AE architecture.

are continuously updated with every additional time step [41]. The outputs $C_{t,i}$ and $h_{t,i}$ from each unit are then passed to all units in the subsequent AE layer. Since the second layer contains fewer LSTM units, the information is progressively compressed, reducing the dimensionality while retaining the most relevant features.

The final LSTM layer in the encoding phase produces highly compressed representations of the multivariate input data [41], forming the latent space, which is denoted as Z. Before the data is passed to the decoder network, a *Repeat Vector* layer is applied to reshape Z to match the length of the input sequence. These repeated vectors then serve as inputs for the decoder network [35].

During the decoding phase, the compressed information is gradually decompressed through successive layers, ultimately restoring the data to its original structure. A dense layer is applied to reshape the reconstructed output to match the original input dimensions [35]. Throughout the training process, the model optimizes its weights and biases to minimize reconstruction error and accurately reconstruct the input multivariate time series sequence [35], [49].

The processing and data flow within an LSTM AE is illustrated in Fig. 5.

Fig. 5 provides a conceptual illustration of the data flow within an LSTM AE. Multivariate input time series x are encoded into a compressed latent representation Z, referred to as the feature space. This latent representation is then decoded to reconstruct the original input sequence, denoted as $\hat{x}$. The figure is intended to convey the general structure and flow of information and does not reflect the exact layer configuration of the implemented model.

V. Early Detection Framework

In this section, we outline our proposed framework for early disturbance detection. We begin with data preparation, followed by the initialization, training, and testing of our LSTM AE model. Finally, we introduce a new evaluation framework. Fig. 6 shows the steps of the process we followed.

A. Data Preparation

As mentioned in Section III-B, the data for all observed sites are contained in spatial and temporal indexed data frames. During preprocessing, the multivariate time series for each pixel was treated individually.

The time series contain a signal for every fifth day. Due to cloud or snow coverage, many data instances are deemed unsuitable and are excluded from the analysis. To filter these instances, we used the SCL layer from the L2A images to retain only pixels classified as either vegetated or nonvegetated. As shown in (11) we decomposed the time series into trend, seasonal component and remainder and applied Tukey's method [50] to the remainder R_t to remove remaining outliers

$$x_t = T_t + S_t + R_t \tag{11}$$

where T_t denotes trend, S_t, seasonal component, and R_t the remainder. We then aggregated the data to weekly values and imputed the gaps to obtain homogeneous time series. First, we smoothed the time series using the Savitzky–Golay filter (SGF) and then filled the gaps with exponentially weighted moving average (EWMA). The SGF smooths and differentiates data by performing a local least squares polynomial fit within a sliding window. It detects the underlying trend of the data while reducing noise. Applying SGF to enhance the quality of satellite image-based time series by smoothing out fluctuations and noise has been widely referenced [51], [52], [53], [54]. A window size is selected, determining the number of observations used in the smoothing process. If the window consists of $2S + 1$ points, the consecutive observations within the window are $x_{i-S}, x_{i-S+1}, \ldots, x_i, \ldots, x_{i+S}$. To model the trend within each window, a polynomial of order r is fitted using least squares regression. The fitted polynomial function

$$F(k) = \sum_{j=0}^{r} \beta_j k^j \tag{12}$$

where k denotes the relative position of a data point within the local window, centered at i, β_j denotes the coefficients of the polynomial function, and r denotes their order. The coefficients of the polynomial function are then calculated using the least squares method by minimizing the following expression:

$$\min \left\{ \sum_{k=-S}^{S} \left(y_{i+k} - F(k) \right)^2 \right\}. \tag{13}$$

The SGF uses all observations, so $x_{i-S}, x_{i-S+1}, \ldots, x_i, \ldots, x_{i+S}$ to determine a new value for x_t at $t = i$.

After smoothing the time series, the gaps were filled using EWMA-imputation, which fills missing values of a time series by calculating the mean from an equal number of observations on either side of a missing value. EWMA employs a weighting factor that decreases exponentially with the number of time steps from the missing value. The weighting factor is given by $w_j = 2^{-j}$, where j represents the temporal distance from the missing value [55].

The LSTM AE model is designed to process sequences of multivariate time series. The length of the input sequences is referred to as the "window size," This defines the observation

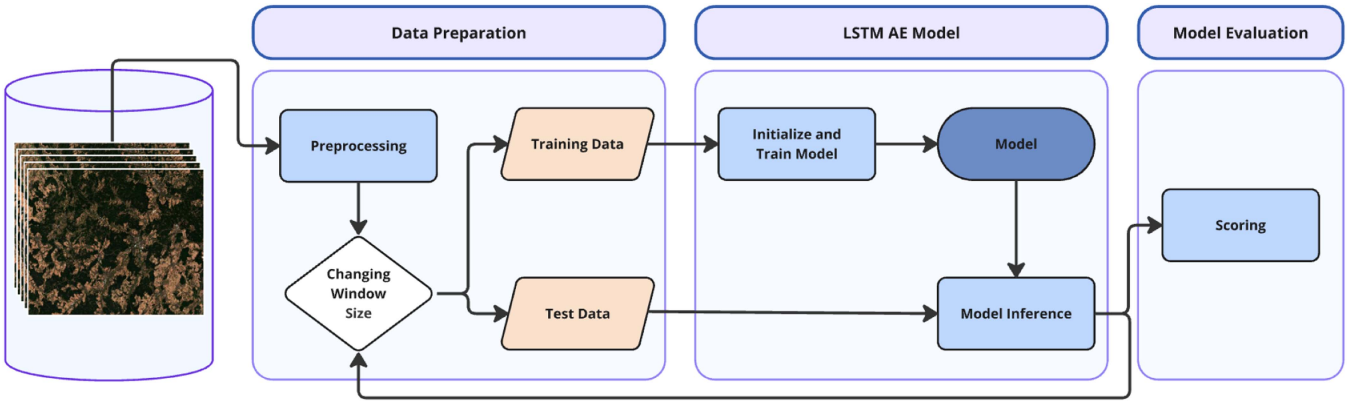


Fig. 6. Methodology outline.

Algorithm 1: LSTM Autoencoder Anomaly Detection.

$Tr \leftarrow$ Training dataset of time series of healthy pixels
$Te \leftarrow$ Test dataset of time series of healthy and unhealthy pixels
$\tau \leftarrow$ Anomaly detection threshold
$ws \leftarrow$ Window size
$x \leftarrow$ time series in Tr or Te
$t \leftarrow$ time step in x
Train LSTM-AE:
for $ws \in \{4, 12, 26, 52\}$ **do**
  for $x \in Tr$ **do**
    $LSTM_AE_{ws} \leftarrow train_model(x)$
  end for
end for
LSTM-AE inference:
for $ws \in \{4, 12, 26, 52\}$ **do**
  for $x \in Te$ **do**
    $\hat{x} \leftarrow LSTM_AE_{ws}(x)$

$$Anomaly_{x,t} = \begin{cases} 1, & \text{if } |x_t - \hat{x}_t| > \tau, \\ 0, & \text{otherwise.} \end{cases}$$

  end for
end for

window of the model and thus the minimum amount of historical data required for the model to operate. For the initial run, we selected a window size of 52 instances (one year) to cover an entire seasonal period. Both, healthy training time series and the testing time series, were preprocessed using the above mentioned methods and the same window size was selected.

B. LSTM AE Model-Training and Inference

Here, we outline the development, training, and evaluation of our model. To assess performance across different temporal contexts, the procedure was carried out for multiple window sizes of 4, 12, 26, and 52 weeks. The complete process is detailed in Algorithm 1.

TABLE I
SENTINEL-2 BANDS, DESCRIPTION, WAVELENGTH RANGE, AND RESOLUTION

Band Number	Band Description	Wavelength Range (nm)	Resolution (m)
B2	Blue	458–523	10
B3	Green	543–578	10
B4	Red	650–680	10
B5	Red-edge 1	698–713	20
B6	Red-edge 2	733–748	20
B7	Red-edge 3	773–793	20
B8	NIR	785–900	10
B11	SWIR I	1565–1655	20
B12	SWIR II	2100–2280	20

NIR = Near infrared; SWIR = Shortwave infrared

a) Initialization: To initialize the model, we constructed a multilayer LSTM AE architecture designed to process multivariate time series sequences derived from Sentinel-2 imagery. As input features, we selected nine spectral bands: B2, B3, B4, B5, B6, B7, B8, B11, and B12. These bands were selected to cover the spectral regions relevant for vegetation monitoring. Priority was given to bands available at 10 m spatial resolution to enhance the model's ability to detect subtle canopy-level changes across forest plots. In spectral regions where no 10 m bands exist, we incorporated the best available alternatives at 20 m resolution in order to capture subtle temporal and spatial variations in forest condition. An overview of the selected bands, their central wavelengths, and spatial resolutions is provided in Table I.

Our model architecture consists of three encoding layers with 256, 128, and 64 LSTM units per layer. A dropout layer with a dropout rate of 0.2 was inserted between the first and second LSTM layer to prevent overfitting of the model [56]. First, a repeat layer was used in the decoder, which repeats the latent space vector to the initial window size. This is followed by two LSTM layers with 64 units and 128 units, as well as a dropout layer with a dropout rate of 0.2. Finally, another LSTM layer with 256 units is inserted into the architecture and a time distributed dense function layer, which compresses the decoder output back to the input size, is used for each time step individually.[4] The multilayer model structure with the number of trainable parameters in each layer is described in Table II.

[4]The model was defined using the open-source ML platform TensorFlow.

TABLE II
LSTM-AE MODEL STRUCTURE

Layer	Output Shape	Parameters
Model Input	$(window_size, 9)$	
Encoder		
256 LSTM Units	$(window_size, 256)$	272,384
Dropout Layer	$(window_size, 256)$	0
128 LSTM Units	$(window_size, 128)$	197,120
64 LSTM Units	(64)	49,408
Repeat Vector	$(window_size, 64)$	0
Decoder		
64 LSTM Units	$(window_size, 64)$	33,024
128 LSTM Units	$(window_size, 128)$	98,816
Dropout Layer	$(window_size, 128)$	0
256 LSTM Units	$(window_size, 256)$	394,240
Time Distribution	$(window_size, 9)$	2,313

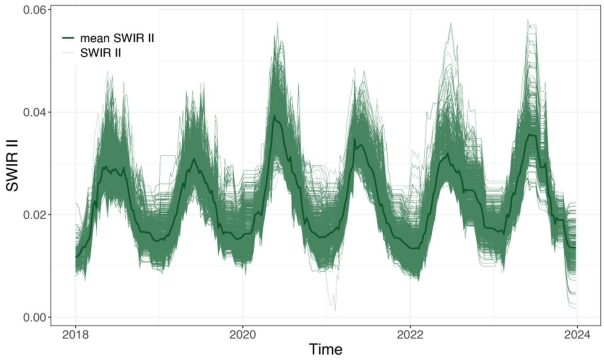


Fig. 7. Short wave infrared II training data samples.

b) Model Training: After defining the model, the preprocessed time series of the healthy training areas, which were described in Section III-A, were visually analyzed, and their health status was assessed using VIs. Pixels exhibiting disturbances at any point were excluded, resulting in 2885 time series from 2018 to 2024. Fig. 7 shows one variable (band) of the training data.

Then, each variable is scaled between 0 and 1 and the time series is segmented according to a selected window size, initially set to 52 to encapsulate a full seasonal cycle. This results in 17 310 batches, each containing 52 time steps and 9 features.

Since the model operates in an unsupervised setting, no labeled anomalies or disturbance classes are required during training. The objective was to learn typical seasonal reflectance patterns of healthy spruce forests, enabling it to accurately reconstruct time series of healthy forests while recognizing anomalous patterns,such as bark-beetle infestations [35], [41].

Each LSTM unit is defined by four input weights (w_f, w_i, w_C, w_o), four recurrent weights (u_f, u_i, u_C, u_o), and four bias values (b_f, b_i, b_C, b_o) [47]. The number of trainable parameters in an LSTM layer is determined by

$$\theta = 4\left(h(h+i)+h\right). \tag{14}$$

The number of units in a layer is denoted as h and the input features are denoted as i. θ represents the tunable parameters [57].

All parameters are optimized during training to best reconstruct normal time series behavior [35], [41]. Fig. 8 describes the operation in abstract terms.

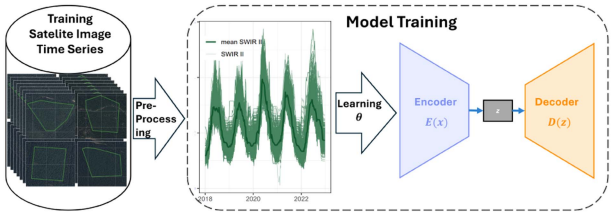


Fig. 8. Process of model training.

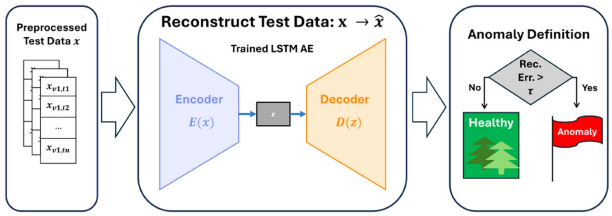


Fig. 9. Process of model inference.

The model was trained using the Keras API, specifically the compile and fit methods. The Adam optimization algorithm [58] was employed for model updates, while mean squared error was used as loss function. The training process involves adjusting the model's parameters – weights and biases – over multiple iterations to minimize reconstruction error. Key training configurations included a batch size of 128, 200 epochs, and a validation split of 0.28. The batch size defined the number of training instances processed before updating parameters, while epochs determined the total number of dataset iterations. This process was repeated for the window sizes of 26, 12, and 4 time steps, resulting in four models.

c) Model Inference: During the inference phase, the models processes time sequences and compresses them into a lower-dimensional representation, known as the latent space. Afterward, the latent space is expanded to match the length of the input sequence, and this representation is then reconstructed by the decoder network [41], [45] (see Fig. 9).

The data transformation within the AE is regulated by the weights of the LSTM units, which are optimized during the training phase. Since these weights are trained to reconstruct healthy time series, the difference between the input x and its reconstruction $\hat{x}$ will be larger if x contains anomalous data. An anomaly at a time step t is identified if the mean absolute error across all features, between $\hat{x}_t$ and x_t exceeds a threshold τ [44].

$$\text{Anomaly}_t = \begin{cases} 1, & \text{if } |x_t - \hat{x}_t| > \tau, \\ 0, & \text{otherwise.} \end{cases} \tag{15}$$

For every model, we reconstruct every multivariate time series within the test dataset. For every time series, each band's time series is reconstructed based on its own temporal pattern, the relationships between different bands, and their temporal dependencies. Note that the model is able to reconstruct some bands better in healthy areas than in unhealthy forest areas, as shown in Fig. 10.

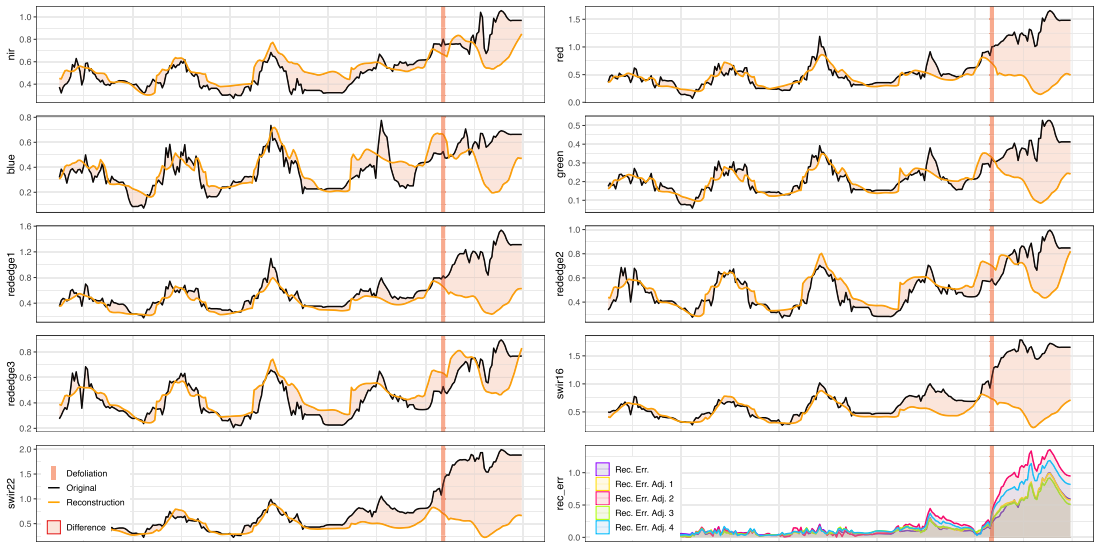


Fig. 10. Original time series and its reconstructions across all features of a single time series, including reconstruction errors calculated using the MAE between different feature combinations. Red vertical lines label the date of defoliation. The applied model is LSTM-AE52.

Consequently, we calculated the adjusted reconstruction errors, which are based on the MAE of different band combinations. This adjustment is intended to help to distinguish between healthy and unhealthy time points. We selected the following combinations:

1) *rec_err:* all bands.
2) *rec_err_adj1:* NIR, RED, REDEDGE 1-3, SWIR I/II.
3) *rec_err_adj2:* REDEDGE 1-3, SWIR II.
4) *rec_err_adj3:* NIR, REDEDGE 1-3, SWIR I, SWIR II.
5) *rec_err_adj4:* NIR, RED, SWIR I, SWIR II.

The threshold was calculated on an individual basis for each window size and band combination. This was achieved by first applying the model to time series representing healthy forest conditions. The 99.8th percentile (quantile 0.998) of the distribution of the resulting reconstruction errors was then selected as the anomaly detection threshold.

All data preprocessing and time series handling as well as training and inference were conducted in R (v4.4.2).[5] The model was trained and evaluated on a CPU-only setup (AMD Ryzen 5 7730 U, 32 GB RAM). Training the LSTM-AE26 and LSTM-AE52 models for 200 epochs required approximately 7.7 and 15.8 h, respectively. Inference on a one-year time series averaged 0.109 s (LSTM-AE26) and 0.094 s (LSTM-AE52) per pixel ($n = 1211$), including preprocessing and reconstruction steps.

C. Evaluation Framework

As mentioned in Section II, in remote sensing-based forest health monitoring, labeled data is extremely difficult to obtain. The exact start of the bark beetle infestation cannot be determined from remote sensing [9]. Therefore, evaluating models that detect early anomalies to predict forest disturbance is very difficult if there is no labeled data. We classified defoliation in pixels based on VI thresholds and created a scoring method that assumes an anomaly window around the defoliation date. The earlier anomalies are detected within the window, the higher they get rewarded.

a) Defoliation Threshold: Several authors used static thresholds based on VIs to determine if spruce vegetation within a cell is defoliated [59], [60]. NDVI is frequently cited in the literature as a health indicator of forest vegetation [59], [60], [61], [62], [63] and is an efficient measure to distinguish dead from healthy spruce trees [60], [61]. Although the 10-m resolution of Sentinel-2 imagery does not allow the analysis of individual trees, changes in crown color, and needle loss alter the NDVI value in a pixel. Previous studies suggest a NDVI range of 0.481–0.584 as threshold to assess defoliation in mixed conifer-deciduous stands [59]. More recent research by Goldbergs and Upenieks [60] evaluated an NDVI threshold of 0.46 to distinguish between living and dead trees within spruce monocultures. Through visual analysis of the images we found 0.53 to be a fitting threshold for the NDVI.

$$\text{NDVI} = \frac{B8 - B4}{B8 + B4}. \tag{16}$$

In addition, we selected the Tasseled Cap Wetness (TCW) index, which is a suitable index for disturbance detection on sentinel time series, especially being more precise during the winter months [11]

$$\begin{aligned}\text{TCW} = {} & 0.1509 \cdot B2 + 0.1973 \cdot B3 + 0.3279 \cdot B4 \\ & + 0.3406 \cdot B8 - 0.7112 \cdot B11 - 0.4572 \cdot B12. \end{aligned} \tag{17}$$

We found a threshold of -0.03 to be especially suited. In our test data, we searched for pixels that fall below the threshold for three consecutive weeks and classified the first instance as defoliation. A portion of the defoliation dates was visually checked and confirmed using Sentinel-2 images.

b) Anomaly Window: Anticipating that the anomaly detection algorithm detects anomalies prior to the apparent defoliated

[5]In addition to standard data science libraries, the following packages were used for geospatial data handling and ML: `terra`, `sf`, `sp`, `keras`, `tensorflow`, `isotree`, `bfast`

state, we implemented a buffer zone to categorize infected cells before the needle loss or discoloration of the canopy (i.e., defoliation). We analyzed the lag between predicted anomaly and defoliation, to generate a buffer period preceding defoliation, during which anomalies are correctly labeled as such rather than erroneously classified as false positives. We chose a 52-week anomaly window to capture early indicators of health decline in spruce stands that lead to defoliation. Bark-beetle infested trees show no evidence of differences in NDVI for up to 75 days after infestation [24], [64]. Timely detection is further hampered by the delayed and irregular vitality response of spruce crowns as well as frequent cloud cover [9], both of which we observed in our time series. Bark-beetles primarily attack trees that are already weakened by stress factors, such as reduced resin or sap flow due to a disturbed water balance [5]. In addition, our method detects general health anomalies – not just bark-beetle infestations – a full seasonal cycle allows us to capture phenological variations that signal tree weakening, improving the detection of true anomalies and enhancing the predictive ability of defoliation events.

If an anomaly is detected within the buffer interval around the defoliation, it is classified as a true positive (TP). If an anomaly is flagged outside the buffer, it is classified as a false positive (FP). If no anomaly is flagged within the buffer, it is classified as a false negative (FN). If a pixel's time series shows no defoliation and no anomaly is flagged, it is classified as a true negative (TN). These results were utilized to calculate the overall accuracy, balanced accuracy, F1 score, commission error and omission error to evaluate the models performance

$$\text{Overall Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \tag{18}$$

$$\text{Balanced Accuracy} = \frac{1}{2}\left(\frac{\text{TP}}{\text{TP} + \text{FN}} + \frac{\text{TN}}{\text{TN} + \text{FP}}\right) \tag{19}$$

$$\text{F1 Score} = \frac{2 \cdot \text{TP}}{2 \cdot \text{TP} + \text{FP} + \text{FN}} \tag{20}$$

$$\text{Commission Error} = \frac{\text{FP}}{\text{TP} + \text{FP}} \tag{21}$$

$$\text{Omission Error} = \frac{\text{FN}}{\text{TP} + \text{FN}}. \tag{22}$$

This methodical approach ensures a robust and transparent evaluation of the models, providing a solid foundation for determining the best-performing model [65].

c) Pre-Defoliation Score: The above measures are commonly used in anomaly detection, but all fail to promote early detection, despite its critical importance in mitigating ecological damage. Kautz et al. [9] highlight that only 23% of the studies on early bark-beetle detection rely on accurate ground truth data, underscoring the rarity of well-labeled datasets. We have developed a novel scoring method to address this gap: the *Predefoliation Score*. It is designed to evaluate and compare anomaly detection models in the context of forest disturbances when no labeled anomaly dates are available. This metric rewards early anomaly detection while penalizing both false positives and late detections, ensuring a more reliable assessment of model performance. (see Fig. 11.)

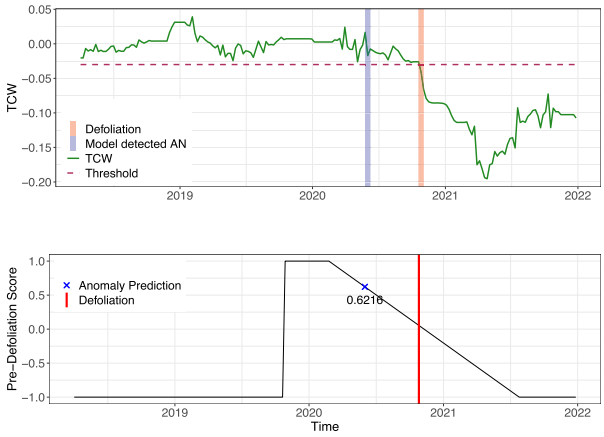


Fig. 11. Predefoliation Scoring system based on the temporal position of anomaly detection relative to defoliation. Early detections are rewarded, while late or false detections are penalized.

To implement the predefoliation Score, we defined an anomaly window around the time of defoliation, i.e., the moment when vegetation loss is certain to have occurred. Anomalies detected earlier within this window receive a higher score, whereas detections following the defoliation event are assigned increasingly negative values. False detections, outside the anomaly window or in healthy time series, receive the lowest possible score (-1). Anomalies detected within the first third of the predefoliation window were assigned a score of 1, as we observed that detections in this time frame provide equally valuable early warnings. The score decreases linearly for later detections. It reaches zero two weeks after the defoliation event. This design accounts for practical constraints, such as cloud cover, which can hinder detection in earlier satellite images. If an anomaly is detected beyond this time frame, it is assigned an increasingly negative score. Anomalies predicted outside the predefined anomaly window or in a healthy forest time series are penalized with a score of -1. Hence, if no anomaly is detected in a time series where no defoliation occurred (i.e., a healthy forest), the model is rewarded with a TN score of 0.5. The predefoliation Score is defined by the following equation:

$$\text{pd_score} = \frac{-3}{2 \cdot k + s} \cdot (\text{lag} - s) \tag{23}$$

where the difference between time of anomaly detection and defoliation time ($t_{\text{anomaly}} - t_{\text{defoliation}}$) is denoted as lag and k represents the anomaly window (52). A shift, denoted as s is added to award an anomaly detected at the time of obvious defoliation with a positive value.

$$\text{pd_score} = \begin{cases} -1, & \text{if } |\text{lag} - s| > \frac{2}{3}k \\ 1, & \text{if } (\text{lag} - s) < -\frac{2}{3}k \wedge \text{lag} >= -k \\ \text{pd_score} & \text{otherwise.} \end{cases} \tag{24}$$

If multiple anomalies were detected within the anomaly window, only the first anomaly was considered, while subsequent

detections were assigned a score of zero to ensure that the models are not rewarded multiple times for a single defoliation event.

The predefoliation score provides a robust framework to evaluate early forest disturbance detection. By prioritizing timely anomaly identification while penalizing false alarms and delayed detections, this metric offers a meaningful way to compare models, even in the absence of perfectly labeled ground-truth data. Future studies can leverage this approach to improve remote sensing-based forest health monitoring.

D. Model Scoring

To address the research objective of analyzing the importance of temporal information, we tested LSTM-AE models using diverse input sequence lengths of 4, 12, 24, and 52 weeks. The adjusted reconstruction errors of all four models were compared with each other.

To assess the performance advantages of our models compared to established methods, we compared the LSTM-AE models to the break detection algorithm BFAST and the other widely used anomaly detection algorithm IF. BFAST detects structural breaks in time series trends, making it particularly suitable for analyzing VIs. We tested the BFAST algorithm on the following VIs: NDVI, DWSI, EVI, and DRS, which are frequently used in bark-beetle detection [15], [16], [19], [66], [67]. BFAST requires a historical reference period for each observed pixel, we set the training phase from January 2018 to January 2020 to ensure a sufficient number of observations. IF is an unsupervised, tree-based anomaly detection algorithm commonly used due to its independence from labeled training data, making it a suitable baseline for comparison with our reconstruction-based approach. We trained IF models on the same dataset as the LSTM-AE models, applying a rolling window strategy with sequence lengths of 26 and 52, as shorter windows (4 and 12) yielded unstable results. Model performance was further optimized by tuning the number of trees, testing configurations of 100, 300, and 500 estimators.

Both algorithms output the timing detected anomalies for each time series, which we then use to compute the predefoliation Score, along with the performance metrics overall accuracy, balanced accuracy, F1-score, commission error, and omission error. Finally, we compared these results to those of the LSTM-AE models to evaluate their relative effectiveness in early anomaly detection.

VI. Results

To evaluate model performance, we applied four LSTM-AE variants using temporal input sequences of 4, 12, 26, and 52 weeks. These were compared against two established anomaly detection approaches: BFAST and IF. In the following, we present the results of these comparisons using the predefoliation score and standard classification metrics.

A. Sequence Length and Adjusted Reconstruction Errors

The results in Fig. 12 show that *rec_err_adj4*, the adjusted reconstruction error based on the NIR, RED, SWIR16, and SWIR22 bands, consistently achieved the highest predefoliation Scores (pd_scores) across all LSTM-AE window sizes.

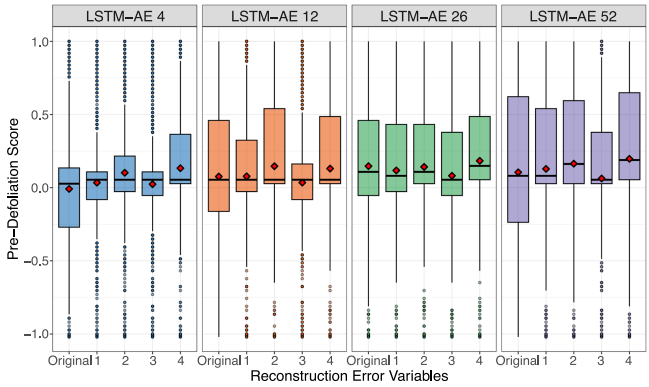


Fig. 12. Predefoliation Score comparison of LSTM models using different sequence lengths and feature combinations used for scoring. Box plots illustrate the distribution of pd_scores for each configuration. While all spectral bands were used during model training, the reconstruction error (MAE) was calculated using different feature combinations to evaluate their impact on anomaly sensitivity.

The pd_score quantifies how early an anomaly was detected relative to a known defoliation event: a score of 0 indicates detection at the time of defoliation (including a two-week buffer), while higher values represent earlier detection. Scores greater than 0.17 correspond to detections made at least one month in advance. Except for window size 12, it yielded the highest mean and median pd_scores, with values of 0.222 and 0.216 for LSTM-AE52, respectively. Consequently, this feature combination was selected for further analysis. TNs were excluded from all box plots to avoid clustering at a pd_score of 0.5.

Furthermore, the window size had a notable effect on model performance. We will refer to these experiments as "LSTM-AE" concatenated with the respective window size (e.g., LSTM-AE26). Models with longer input sequences (LSTM-AE26 and LSTM-AE52) exhibited higher median and mean pd_scores. For instance, LSTM-AE26 achieved a mean pd_score of 0.194, while LSTM-AE4 yielded a lower value of 0.135. Although LSTM-AE4 produced fewer false positives and late detections and had lower score variance, it underperformed in early anomaly detection.

Fig. 13(a) displays the pd_score distributions as box plots, showing that pd_scores increase with sequence length. LSTM-AE52 produced the highest values overall. Fig. 13(b) presents cumulative probability curves, highlighting that LSTM-AE26 and LSTM-AE52 detect a greater proportion of anomalies with scores above 0.17, indicating earlier detections relative to defoliation.

B. Performance Comparison Against Baseline Models

To compare the performance of our LSTM-AE models to BFAST and IF, we calculated the mean predefoliation score (PDS_{μ}) and the percentage of pixels where anomalies were flagged more than four weeks prior to defoliation (*PDS %*). We additionally computed overall accuracy, balanced accuracy,

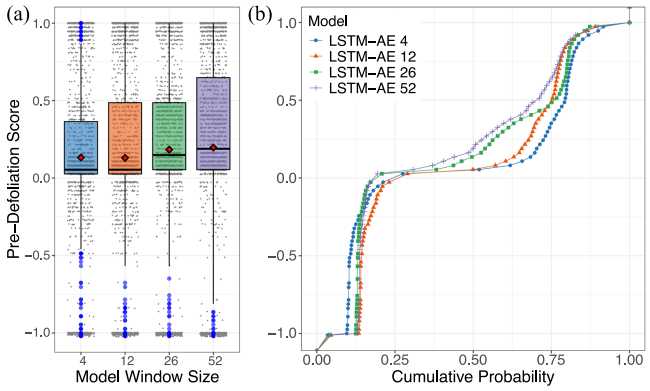


Fig. 13. Predefoliation Score comparison using different sequence lengths in LSTM models. (a) Box plots illustrating the distribution of scores across different models. (b) Cumulative probability curves depicting the score distributions. Higher scores indicate earlier detections relative to defoliation.

TABLE III
PERFORMANCE SUMMARY OF THE LSTM-AE MODELS AND THE COMPARISON MODELS BFAST AND IF

Model	PDS_μ	PDS %	QA	BA	F1	OE	CE
LSTM-AE4	0.180	40.45	0.811	0.821	0.876	0.194	**0.041**
LSTM-AE12	0.221	52.37	0.823	0.839	0.863	0.202	0.060
LSTM-AE26	0.265	62.03	**0.866**	**0.873**	**0.899**	**0.146**	0.051
LSTM-AE52	**0.274**	**64.84**	0.858	0.872	0.891	0.165	0.046
BFAST-DRS	0.197	58.05	0.741	0.723	0.807	0.233	0.148
BFAST-DWSI	0.076	54.89	0.710	0.655	0.792	0.195	0.221
BFAST-EVI2	-0.189	38.11	0.487	0.531	0.519	0.592	0.287
BFAST-NDVI	0.080	51.64	0.630	0.644	0.690	0.394	0.200
IF-W26	0.003	42.99	0.479	0.544	0.479	0.644	0.267
IF-W52	-0.098	36.60	0.431	0.501	0.418	0.697	0.323

F1 score, commission error, and omission error to comprehensively assess model performance. We selected these performance metrics to ensure comparability with other studies, such as Schiller et al. [29], which employed similar evaluation criteria for classification-based disturbance detection. Since BFAST requires a two-year reference period, we excluded all pixel time series with defoliation events occurring before 2020 from the evaluation. The performance results are summarized in Table III.

Table III summarizes model performance using rec_err_adj4, based on Sentinel-2 bands NIR, RED, SWIR16, and SWIR22. Results are shown for LSTM-AE models with input windows of 4, 12, 26, and 52 weeks; for BFAST using multiple vegetation indices (DRS, DWSI, NDVI, EVI2); and for IF applied with input windows of 26 and 52 weeks. The metrics used for comparison are PDS_μ, *PDS%*, balanced accuracy (BA), F1 score, commission error (CE), and omission error (OE). Among the LSTM-AE models, LSTM-AE4 had the lowest commission error (0.041), while LSTM-AE26 showed the highest balanced accuracy (0.873) and F1 score (0.899). LSTM-AE52 reached the highest values in PDS_μ (0.274) and *PDS%* (64.84%). The baseline models BFAST and IF showed lower scores across all metrics. The distribution of predefoliation scores across the LSTM-AE, BFAST, and IF models is illustrated in Fig. 14.

Fig. 14 shows the distribution of Pre-Defoliation Scores across all models. LSTM-AE models tend to produce higher scores as window size increases. BFAST models display greater variance, and the DRS variant yields a positive median. IF models show lower medians and narrower interquartile ranges.

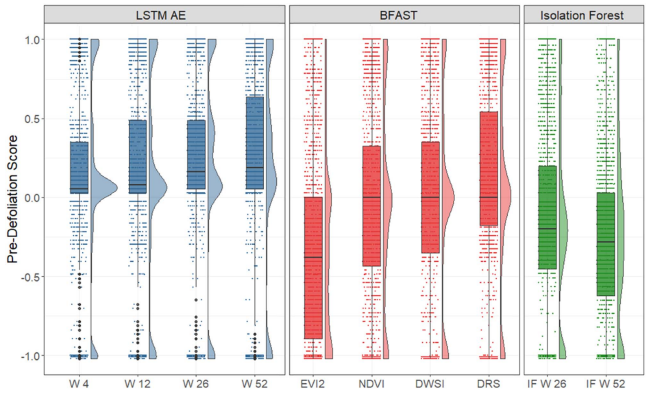


Fig. 14. Distribution of predefoliation scores across all models. LSTM-AE models are shown in blue, BFAST variants in red and IF in green. Box plots and violin plots illustrate the spread, density, and median pd_scores for each method.

C. Evaluation Maps

The predefoliation scoring system enables the creation of maps that visualize model performance within an observed area. Fig. 15 visualizes the predefoliation scores of the respective pixels in a larger spruce cultivation area for the time span from 2020 to the end of 2024.

Both LSTM-AE variants demonstrate more robust anomaly detection, with significantly fewer false positives than BFAST. This is visualized in the test area shown in Fig. 15, comprising approximately 44 500 pixels (445 ha). LSTM-AE26 produces around 6500 false positives and LSTM-AE52 about 8000, compared to approximately 10 500 for BFAST. LSTM-AE26 also results in slightly fewer late detections (red pixels) than LSTM-AE52, by roughly 300 pixels. However, LSTM-AE52 achieves the highest number of early detections, defined as anomalies detected more than four weeks prior to defoliation (pd_score $>$ 0.17).

VII. DISCUSSION

This study demonstrates the effectiveness of the unsupervised deep learning LSTM-AE model, for early detection of forest disturbances in Norway spruce stands using multispectral time series. By focusing on physiologically relevant spectral bands (NIR, RED, SWIR16, and SWIR22) and optimizing sequence lengths, the LSTM-AE models, especially the 26-week variant, achieved a strong balance between early anomaly detection, low commission error and small input data demands. Compared to traditional methods like BFAST and IF, our models consistently outperformed across multiple evaluation metrics. These findings highlight the potential of our model for early anomaly detection in operational forest health monitoring and proactive forest management, particularly in scenarios where labeled data are limited.

A. Spectral Sensitivity and Temporal Resolution

As shown in Fig. 12, the most effective band combination for anomaly scoring was *rec_err_adj4*, consisting of NIR, RED, SWIR16, and SWIR22. These bands yield great sensitivity to

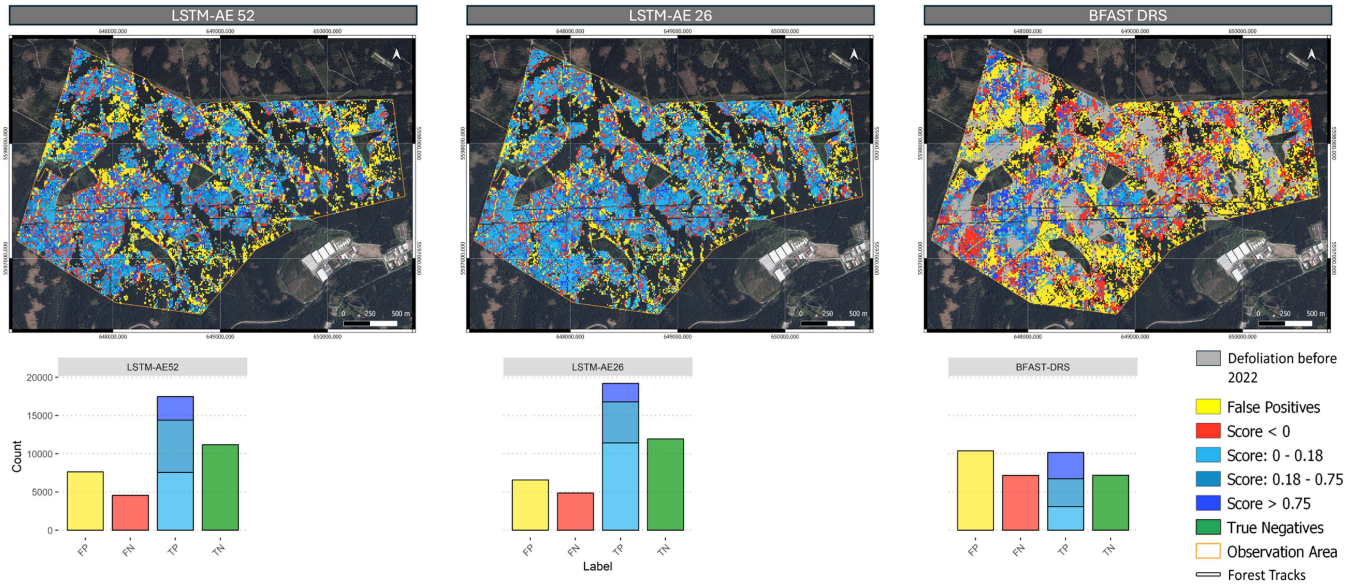


Fig. 15. Predefoliation score maps for LSTM-AE52, LSTM-AE26, and BFAST over a spruce cultivation area in the Thuringian Forest. The pd_score values indicate the timing of detected anomalies relative to the defoliation event. Yellow pixels represent false positives; red pixels indicate late detections. Blue pixels denote timely or early detections: medium blue corresponds to detections more than four weeks prior to defoliation (pd_score ≥ 0.18), while dark blue indicates detections more than 26 weeks prior (pd_score > 0.75).

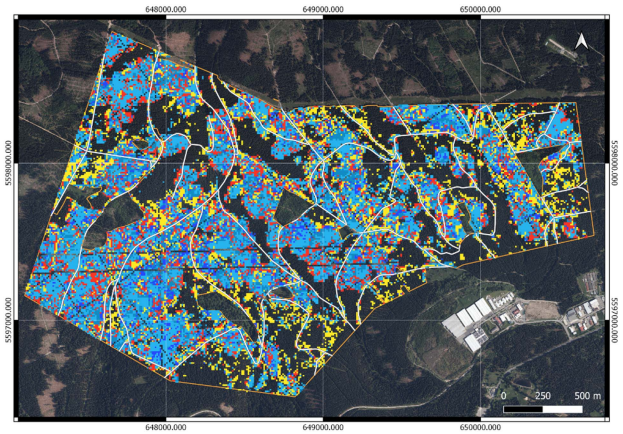


Fig. 16. LSTM-AE26 score map with forest tracks. Forest tracks are overlaid on the evaluation map, revealing that many false positives predicted by the LSTM-AE model align with these paths.

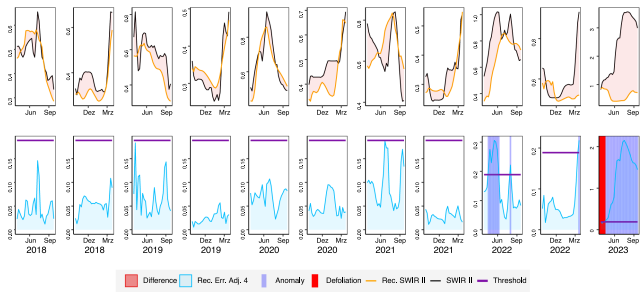


Fig. 17. Anomaly detection in short time series using LSTM-AE26. The figure demonstrates the model's capability to detect anomalies early despite using only short input sequences. The upper row shows the original and reconstructed values for the SWIR22 band, while the lower row displays the corresponding reconstruction error across the rec_err_adj4 features. LSTM-AE26 is applied independently to each input window.

vegetation-related changes as SWIR relates to water content, NIR reflects mesophyll structure, and RED is sensitive to chlorophyll content in vegetation [68]. Furthermore, we noticed that these bands tend to deviate earlier or more distinctly from normal temporal patterns under disturbance conditions. Although the model still utilizes all band inputs for reconstruction, restricting the reconstruction error calculations to these bands enhanced sensitivity to vegetation changes and allowed earlier detection of disturbances. Using other combinations or all bands could dilute or delay the anomaly signal.

However, it is important to note, remotely detectable changes in canopy vitality often lag behind the actual infestation event. For example, bark beetle infestations in late summer may not cause visible spectral changes until the following spring due to winter dormancy [9]. To address this, our anomaly scoring uses a 52-week window to ensure delayed mortality signals are not misclassified as false positives.

The chosen sequence length of a model determines how temporal relationships between features are captured. Smaller windows (e.g., size 4) focus stronger on short-term dependencies and relationships between features. Larger windows capture broader temporal patterns, enabling more abstract representations of the time series. However, this can lead to stronger deviations between the reconstructed and original signals, increasing the risk of false positives when short-term phenological changes are mistakenly flagged as anomalies. Although LSTM-AE52 excels in early detection, LSTM-AE26 offered the best tradeoff, combining strong early detection with minimal false alarms. This tradeoff illustrates how window size should be tuned based on operational needs and tolerance for false positives.

In a similar vein, the interpretation of anomaly scores and the classification of a time step as anomalous depend strongly on the selected threshold. In this study, we used a strict threshold

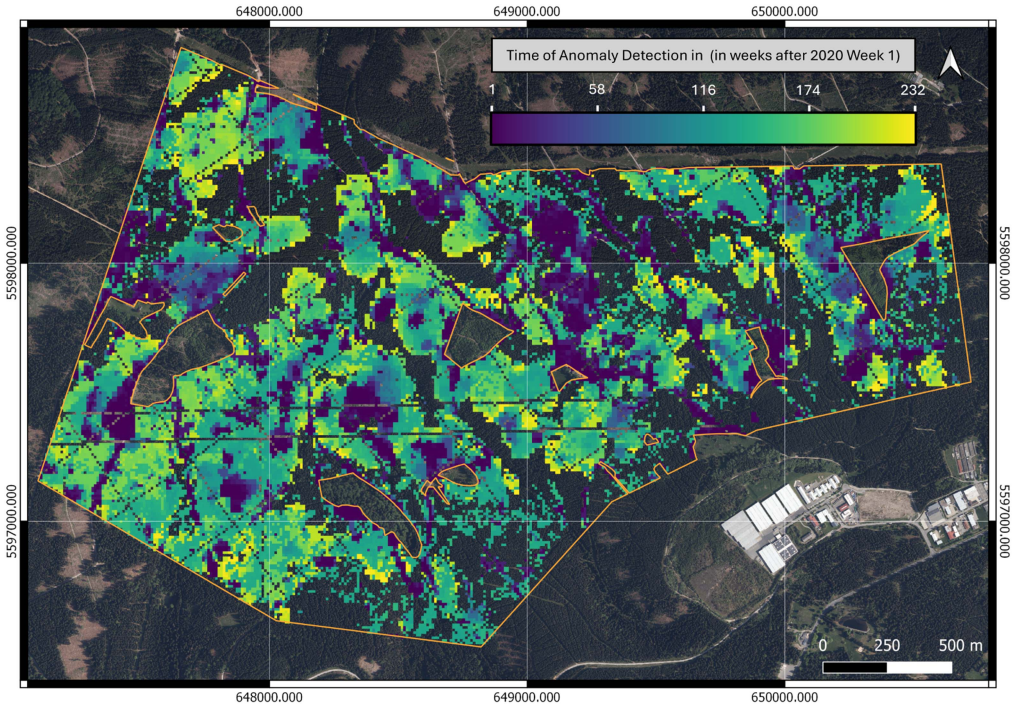


Fig. 18. Anomaly detection map. The gradient change symbolizes the weeks after the beginning of the observation, in which anomalies are detected. The used model was LSTM-AE26.

corresponding to the 0.998 quantile of the anomaly score distribution in a healthy test area. This allows the detection of subtle deviations from normal behavior, enabling early disturbance detection but also increasing the risk of false positives. The threshold can be adjusted depending on the application context.

Compared to the baseline models, the LSTM-AE models demonstrated superior performance across all evaluation metrics. BFAST using the DRS index was the strongest among them but still lagged behind both LSTM-AE models. These findings highlight the advantage of deep learning models that incorporate temporal contexts, particularly for early stage detection of defoliation. As shown in Fig. 14, the LSTM-AE models consistently produce higher predefoliation scores, indicating stronger early detection capabilities. In contrast, IF and most BFAST variants show lower or more variable scores, suggesting a more reactive detection behavior.

The evaluation maps (see Fig. 15) confirm this trend, where BFAST also shows substantially more false positives and less early detections. Many FPs in the LSTM-AE models match forest tracks (see Fig. 16). These false positives correspond to nonforest features, suggesting the need for improved masking.

B. Implication on Forest Monitoring

Unlike more complex architectures such as Transformers, which may require millions of labeled samples [29], our approach achieved strong performance using just 2885 pixel time series. Moreover, our model is suited for, wall-to-wall monitoring of spruce forests at federal state to national scales, given its short time series inputs. An example of the model's ability to detect anomalies before the defoliation on short input time series, by analyzing the temporal relationships between all variables, is illustrated in Fig. 17.

Furthermore, the LSTM-AE models can be used to generate anomaly detection maps, which provide a spatial and temporal representation of detected anomalies within a given area. This visualization allows for a more intuitive interpretation of when and where disturbances in spruce health occur over time. Retrospective application of the anomaly detection method to previously disturbed forest regions enables a deeper understanding of the patterns of progression of disturbances. Fig. 18 visualizes the disturbance dynamic, in one of the test areas, showing anomalies that initially appear in central hotspots and then expand outward. Despite being tested on a small area, the approach holds significant potential for investigating disturbance dynamics on a broader scale. The analysis of breakout patterns has become a key focus of research with important implications for forest management planning [69], [70], [71].

To assess the generalizability of the model, it should be applied to a broader range of study areas that include diverse forest structures and tree species. Furthermore, transfer learning methods may offer a promising approach for adapting the model to new regions and conditions without extensive training. While this study provides an important foundation, further research is required to explore and validate the model's applicability across different ecological settings.

C. Limitations

While the LSTM-AE models demonstrated strong performance, several limitations should be acknowledged. First, inference time was longer than for baseline methods when using

long input sequences (e.g., six years), which may constrain long-term, large-area reconstructions in computationally limited environments. However, this was partly due to the absence of GPU acceleration and can be mitigated with appropriate hardware. Second, as an unsupervised method, the LSTM-AE lacks direct interpretability. The model does not provide insights into which spectral bands or temporal features contribute most to a detected anomaly. Although not explored in this study, techniques for evaluating feature importance and improving model interpretability such as integrated gradients [72] should be considered in future research. Third, this study focused on relatively homogeneous Norway spruce stands within a limited geographic region. While this constraint limits immediate generalizability to other forest types or biomes, its focus lies on the pressing issue of spruce decline due to bark beetle outbreaks. Our study establishes a methodological foundation for early disturbance detection. Future research will aim to adapt and test the model across more diverse forest ecosystems, using approaches such as transfer learning to support broader applicability.

VIII. Conclusion

We introduced an anomaly detection framework for monitoring forest health dynamics in spruce-dominated stands, using bark beetle outbreaks as a case study. The approach is based on an unsupervised deep learning algorithm, specifically, a multilayer LSTM Autoencoder, trained on Sentinel-2 time series to identify deviations from typical spectral–temporal patterns. Although not explicitly trained to detect specific disturbance types, the model was applied to pure spruce stands where canopy deterioration is strongly associated with Ips typographus outbreaks. Thus, the detected anomalies likely reflect early bark beetle-induced stress. More generally, the model will also respond to other disturbances, such as storm damage, drought-induced dieback, or clear cuts, provided they cause detectable spectral changes over time.

The model was tested with different input window sizes, revealing that larger windows lead to earlier detections of disturbances, allowing the identification of early signs of stress prior to visible damage. Notably, the LSTM-AE52 variant achieved an accuracy of 87%, with 65% of the detected anomalies flagged more than four weeks before visible defoliation occurred in the pixel. In addition, we found that a window size of 26 time steps was sufficient to ensure robust and early anomaly detection. The memory-efficient design of our model eliminates the need for labeled training data or extensive historical records for each pixel. This renders it well-suited for large-scale near-real-time monitoring, where older time inputs can be ignored. By capturing both spectral and temporal relationships across all Sentinel-2 bands, our method provides a powerful tool for detecting subtle changes in the health of pure spruce stands, that can indicate early signs of stress, due to abiotic or biotic factors such as drought, or insect attacks. These early signs of stress can serve as predictors of subsequent defoliation events.

The proposed models were evaluated against BFAST and the IF algorithm. The results of this comparison highlight the superiority of the LSTM Autoencoder approach, particularly in terms of early anomaly detection. Furthermore, we introduced a novel scoring method that complements conventional performance metrics by rewarding early detections, a crucial factor in disturbance prediction tasks.

This work provides a foundation for future research by demonstrating the potential of LSTM Autoencoders for early detection of forest disturbances based on Sentinel-2 time series. The model and evaluation framework developed in this study were tailored to the specific context of detecting early signs of disturbance in pure conifer (Norway spruce) stands in Thuringia. While this focused approach ensured a controlled and consistent test environment, it naturally limits the generalizability of our findings. In future work, we aim to extend our analysis to broader spatial domains, incorporating diverse forest types such as deciduous and mixed stands, and applying the approach to additional geographic regions within Central Europe. This will allow for a more comprehensive assessment of the model's robustness and general applicability across varying forest structures and landscapes. Furthermore, we aim to enhance the interpretability of the model by providing greater transparency regarding feature importance. Ultimately, the approach presented here provides a flexible foundation for developing scalable early warning systems for forest health and holds potential for transfer to other domains of environmental anomaly detection.

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